

HINATUAN, Philippines

WMO: 987550

Lat: 8.375N

Lon: 126.318E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.6	73.6	70.0	110.2	75.8	71.0	114.3	76.5	18.7	80.7	14.4	79.3	3.2	270	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	92.7	81.2	91.7	81.5	90.8	81.5	83.5	90.1	82.9	89.5	82.5	89.1	6.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.6	164.6	89.0	81.0	161.0	88.4	80.6	158.7	88.0	47.6	90.5	46.9	89.8	46.4	89.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.9	10.5	69.8	95.7	3.4	1.2	67.4	96.6	65.4	97.3	63.5	98.0	61.1	98.9		
(5)			68.4	84.9	3.1	1.0	66.1	85.6	64.3	86.3	62.5	86.8	60.3	87.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	79.7	79.8	80.5	81.8	82.8	82.8	82.8	83.4	83.0	82.6	81.3	80.8		
	DBStd	2.19	2.11	1.92	1.68	1.64	1.70	1.70	1.83	1.79	1.68	1.70	1.90	1.74		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11608	919	835	946	954	1018	985	1017	1037	991	1010	940	956		
	CDD65	6133	454	415	481	504	553	535	552	572	541	545	490	491		
	CDH74	64250	3916	3661	4683	5484	6383	6038	6078	6448	5979	5907	4927	4746		
	CDH80	22050	871	882	1311	1857	2423	2332	2361	2655	2358	2189	1520	1291		
(14) Wind	WSAvg	4.0	4.2	4.3	4.2	3.7	3.6	3.6	3.9	4.3	4.0	3.9	3.5	4.2		
(15) Precipitation	PrecAvg	175.3	28.5	23.1	17.6	13.8	10.9	9.9	7.9	7.8	8.1	8.9	13.7	24.1		
	PrecMax	233.3	61.0	51.5	29.3	33.0	20.5	17.2	13.5	14.2	17.2	19.7	29.2	38.4		
	PrecMin	122.8	7.0	5.7	4.8	4.7	4.1	3.1	3.2	1.6	3.4	2.4	0.7	6.0		
	PrecStd	31.7	13.0	12.0	6.7	7.2	4.4	4.3	2.6	3.5	3.5	4.4	6.4	8.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.8	89.0	89.9	91.3	92.5	92.7	93.4	94.4	94.1	93.4	91.3	90.4		
		MCWB	80.8	81.0	81.3	82.2	82.5	82.7	81.0	80.1	81.0	80.6	81.6	81.9		
	2%	DB	87.2	87.6	88.5	89.9	90.9	91.2	91.7	92.4	91.9	91.4	89.8	88.6		
		MCWB	80.1	80.5	80.7	81.7	82.2	82.4	81.2	80.7	81.2	81.1	81.6	81.1		
	5%	DB	85.9	86.3	87.2	88.8	89.7	90.1	90.3	90.9	90.6	89.8	88.5	87.1		
		MCWB	79.4	79.8	80.0	81.2	81.9	82.0	81.1	81.0	81.2	81.1	81.1	80.3		
	10%	DB	84.5	85.0	85.8	87.5	88.5	88.9	89.1	89.7	89.3	88.3	86.9	85.7		
		MCWB	78.8	79.0	79.2	80.6	81.4	81.5	80.8	80.9	81.0	80.6	80.4	79.8		
	0.4%	WB	81.8	82.2	82.6	83.4	84.0	84.3	83.4	83.5	83.7	83.5	83.4	83.1		
		MCDB	87.1	87.6	88.6	89.8	90.4	91.1	90.4	90.3	90.7	90.1	89.3	89.0		
	2%	WB	80.8	81.1	81.4	82.4	83.1	83.2	82.5	82.6	82.8	82.4	82.4	81.8		
		MCDB	85.9	86.5	87.1	88.8	89.5	90.0	89.7	89.7	89.8	89.2	88.7	87.3		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	80.0	80.4	80.6	81.6	82.4	82.5	81.8	81.8	82.0	81.6	81.5	80.9		
		MCDB	84.8	85.6	86.1	87.7	88.7	89.2	88.8	89.1	89.0	88.2	87.3	86.0		
	10%	WB	79.2	79.4	79.8	80.9	81.7	81.7	81.0	81.1	81.2	80.8	80.7	80.2		
		MCDB	83.5	84.0	84.9	86.6	87.6	88.1	87.9	88.4	88.0	87.3	86.0	84.8		
		MDBR	9.0	9.7	10.3	11.0	11.6	12.4	12.8	13.7	13.4	12.7	10.7	9.8		
		MCDBR	11.5	12.0	12.7	13.1	13.6	14.3	14.7	15.3	15.4	14.7	13.5	12.2		
(36) Mean Daily Temperature Range	5% DB	MCWBR	6.0	6.5	6.6	6.7	6.9	7.5	7.2	7.3	7.5	7.1	7.0	6.4		
		MCDBR	10.5	11.2	11.6	12.4	13.0	13.8	13.9	14.5	14.5	13.7	12.5	11.3		
	5% WB	MCWBR	6.0	6.5	6.6	6.9	7.0	7.7	7.6	7.8	7.9	7.5	7.1	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.454	0.442	0.426	0.418	0.408	0.397	0.406	0.427	0.425	0.436	0.419	0.437		
	taud		2.346	2.380	2.436	2.457	2.499	2.539	2.502	2.424	2.420	2.394	2.470	2.409		
	Ebn at Noon		273	280	285	283	280	279	278	277	280	278	282	275		
	Edh at Noon		40	40	38	37	35	33	34	37	38	39	35	37		
(44) All-Sky Solar Radiation	RadAvg		1231	1440	1629	1724	1686	1569	1576	1647	1662	1581	1362	1280		
	RadStd		144	153	129	131	113	102	109	99	130	111	126	114		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page